

The Ephemeris

July 2013

Volume 24 Number 7 - The Official Publication of the San Jose Astronomical Association.



Houge Park June Events

7/7

Solar observing. 2 - 4 p.m.
Fix-It Day 2 - 4 p.m.

7/12

Star party. 9:30 p.m. - midnight.

7/20

6 - 7:30 p.m. Board of Directors
7:30 - 8 p.m. General Meeting
8 - 10 p.m. Movie Night.

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Treasurer: Michael Packer
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From the Editor - Mina Wagner

**Oh, the places you'll go...
to observe from a dark sky!**

Hello fellow sky lovers! By this time many of you will have gone to a dark site to observe the wonders of our Universe. If you haven't...WHAT are you waiting for? The longest day of the year has come and gone which means that winter is just around the corner (nooooooooooooo!!!)

If you have gone, tell us a bit about it. Where have you gone? How far away did you go? Describe your observing area. Is there a shelter? Are there any restrooms? Do you sleep over? Do you camp or do you have a comfy bed? How late do you observe? Do you stop somewhere to get some goodies for midnight? Until what time do you observe? Do you observe with friends or are you a lone wolf? How was the weather? Was it cold? Did you see any wildlife? Do you log your observations? Do you sketch what you see? Do you use an electronic Star Chart? Do you spend the night? What did you observe?

I'm just curious. Maybe some people would be convinced of going out to observe by hearing others' experiences. Do drop me a line with details of your outings to: reyes.mina@gmail.com

In this edition you'll enjoy our usual great article by Akkana Peck, Michael Packer, Paul Kohlmliller, Mark Wagner and all the information about what goes on at the SJAA. You should join us. It's fun!

SJAA Movie Night



The SJAA will hold its first Movie Night at the Saturday, 20 July General Meeting. We asked for suggestions on movie titles, and thank you to all who sent in ideas. These were the top three:

- Forbidden Planet
- Galaxy Quest
- Star Wars Episode IV: A New Hope

The winner: Hidden Planet

Please be aware that the movie will take the place of a guest speaker for the 20 July general meeting.

Acknowledging our printer

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View The Northern Lights In California? Yes!

Michael Packer

The northern lights, aurora borealis, or simply aurora is something Californians usually think of as occurring in, well, the north: Greenland, Iceland, Alaska and Canada. These areas are famous for dazzling displays of red, green, orange and deep blue lights, especially in the winter months. But can you see an aurora in California and is it worth tracking down this far south of the pole?

Anyone can tell you what aurorae (or auroras) look like and you can see plenty of images online. But until you've seen this celestial light show unfold yourself you can't know the allure and marvel. The Aurora borealis is aptly named after the Roman goddess of dawn. When I lived in the east coast I got my first view of the northern lights during the early morning peak of the 2000 Perseid meteor shower. The meteor shower was a bust, however the dawn of color from the aurora changed the way I look toward northern horizon forever. When Dirk Obudzinski (San Francisco, CA) saw his first aurora way up in Fairbanks Alaska, it changed his life forever. Since 2001 he has professionally photographed aurorae at some of the most spectacular locations in North America: the Arctic Circle, Yellowstone National Park, the mesas of Arizona and, you got it, California.

To be sure, it would be great to catch your first aurora close to the north or south magnetic poles as possible. This is where charged particles from the Sun converge as they follow Earth's magnetic field to its strongest points of origin and light up the sky. By the way, aurora in the southern hemisphere is called the Southern Lights or Aurora Australis.



Mt Shasta July 15, 2012: Northern Lights with Pleiades. Image captured at 03:05 am (PDT) during a major geomagnetic storm associated with an X1.4 class flare and resulting CME that left the Sun on July 12th. Canon EOS 5D MKII - Lens: Canon EF 24mm f/1.4 L II - 9 sec. exposure. © Dirk Obudzinski <http://borealis2000.com/>



Aurora Borealis from ISS over Canada January 2012 shows solar particles concentrated over the north magnetic pole and lighting up the sky.

But if you want to see or image an aurora without the travel price tag you need not leave this state. Since 2001, Dirk has driven outside the Bay Area's sky glow and successfully imaged the northern lights at Mount Shasta, Napa, Lake Berryessa, outside Oakdale and east of Gilroy at the San Luis Reservoir. (continued on next page)

(continued: View The Northern Lights In California? Yes!)



Aurora April 17 2001 near Oakdale (Left) and San Luis Reservoir October 29, 2003 (Right) © Dirk Obudzinski



Lake Berryessa July 27, 2004 © Dirk Obudzinski

These shots were taken by before Dirk gave up film photography. The equipment he used was Nikon EM w/50mm 1.4 lens and Canon AV-1 w/50mm 1.4 lens. Exposures were in the range of f/1.4-2.0 and 3-12 seconds depending on aurora's brightness et cetera.

A brief record search of SJAA and TAC online archives came up with nothing on viewing or imaging aurora locally. I asked SJAA's veteran astronomer Jim Van Nuland about past members who talked about viewing Aurora. Here's what he said:

“Several of us were at Houge Park [2001]. At about 11 pm, we suspected a pink glow to the northeast. That's the direction over the Camden shopping center, and it's usually yellow. Comparing with other directions supported the pinkness. It was generally circular, and extended from NNE to ENE, roughly 45 degrees in elevation and azimuth. I went home. In my back yard a few blocks northeast of Houge, I was confident that the northeast sky was faintly pink. I e-mailed a sister who lives in mid-Wisconsin, well away from cities. Within half an hour, she replied that her sky was on fire.”

Want to learn:

How is the Aurora made?
Why there are colors in Aurora?
Aurorea on Jupiter?
Aurora In Folklore?

Visit Christina Shaw's page at the University of Alaska Fairbanks
<http://lnk.nu/ffden-2.phys.uaf.edu/2ggj.htm>

3 Steps On How to Catch an Aurora

Jim saw one of the biggest Aurora events so far this century in 2001. And his remarks “faint pink glow” compared to his sister's “sky on fire” should tell you that viewing aurora this far south can be a difficult task. These steps can help.

Step 1: Ignore all the next steps and just sign up for Dirk Obudzinski's Aurora Alert for middle and low latitudes: <http://borealis2000.com/newsletter.html>. Dirk doesn't email out an alert unless there is a high degree of confidence, Kp values of at least 6 or 7 (for an explanation of Kp Values please read Step 3). He hasn't sent anything out for more than a year. This does not mean there aren't any aurora to be had. He's recently snapped some nice ones. He just doesn't want to be nailed for sending folks out into the boondocks for a miss. NOAA also offer this service and you can chose what Kp value, 1 through 9, you would like to have an email alert sent. Sign up at <http://lnk.nu/pss.swpc.noaa.gov/2ggo.aspx>. That's very nice. What's nice about Dirk's site is you have the advantage of a California local with years of experience.

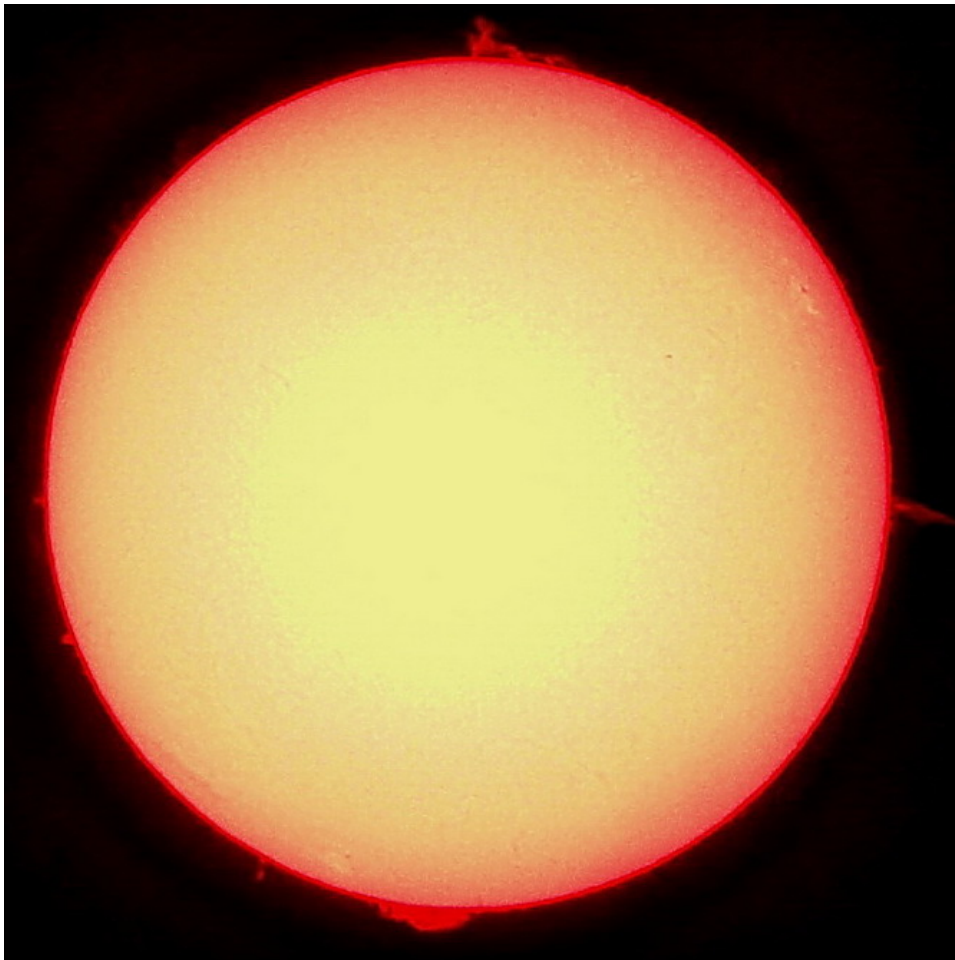
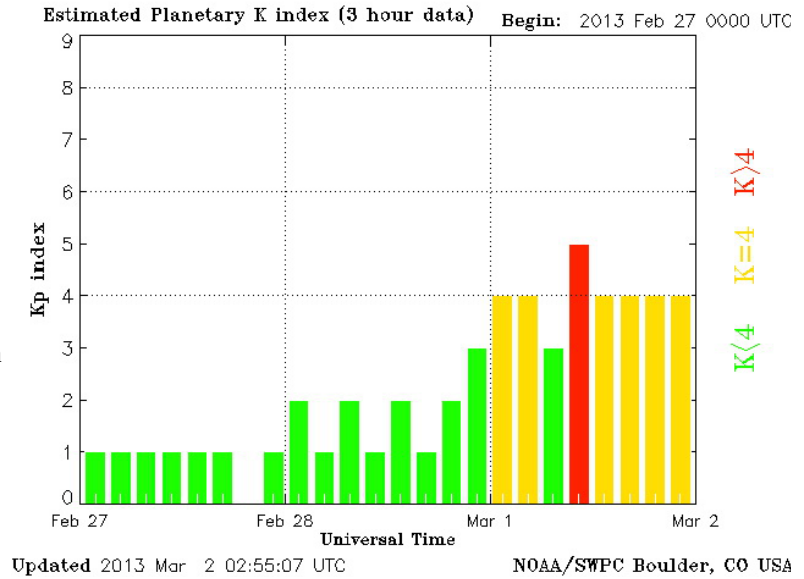
To further increase your chances follow the steps below. Dirk catches aurora 50% of the time he ventures out into the night. And if you compare 50% to the percent chance of hunting down a comet, super nova, or flash from a lunar impact, that's good celestial gravy right there folks! (continued on next page)

(continued: View The Northern Lights In California? Yes!)

Step 2: Stake out a dark sky location (suitable for faint NGC galaxy hunting) with a good low view of the northern horizon and with little or no sky-glow from a city. Light Pollution at your back is OK. Dirt's likes (although he is always looking) Lake Berryessa. You will also need to know this location's northerly skyline well. Illuminated clouds, fog, dust or haze can trick an unseasoned aurora chaser into thinking there's an aurora when it's not. Aurora in California can be fainter than giegenshiesn or zodiacal light but yield beautiful images.

Step 3: Monitor space weather prediction sites (example below) and monitor the Kp Index:

The Kp Index is a scale from 1 to 9 used to measure the size of a geomagnetic storm (9 is most severe). Kp indicates how strong the Earth's magnetic field fluctuations are due to an incoming geomagnetic storm - or a temporary disturbance of the Earth's magnetosphere from the solar wind. Kp index is actually a weighted average of K-indices from a network of geomagnetic observatories located throughout the hemisphere. I found that one of these very few geomagnetic observatories is in California, FRN Fresno. So if we can get real time data from this station, bay area astronomers would have exceptional localized info on aurora being seen in our area. Some good sites to monitor space weather are www.spaceweather.com and Dirk's site. And here is good page to monitor the Kp index <http://www.swpc.noaa.gov/alerts/k-index.html>. The below graph was taken from this site and shows Kp > 4 in March 1st 2013. With data like this available today on the net, amateur astronomers in California have a good chance to catch the Northern Lights!



El Sol on Fathers Day



Have a day in the Sun
at the SJAA's Solar Sunday.
First Sunday 2-4 pm each month.



Shallow Sky Mad Moon Models

By Akkana Peck

Last month I stumbled upon a loony moon book I hadn't seen before, one that deserves consideration by all lunar observers.

The book is *The Moon: Considered as a Planet, a World, and a Satellite* by James Nasmyth, C.E. and James Carpenter, F.R.A.S. It's subtitled "with twenty-six illustrative plates of lunar objects, phenomena, and scenery; numerous woodcuts &c." It was written in 1885.

You may recognize the name Nasmyth: his name is attached to a modified Cassegrain focus design used in a lot of big observatory telescopes. Astronomy was just a hobby for him, though; he was primarily a mechanical engineer. His coauthor, James Carpenter, was an astronomer at the Royal Greenwich Observatory.

The most interesting thing about their book is the plates illustrating lunar features. In 1885, photography wasn't far enough along to get good close-up photos of the moon through a telescope. But Nasmyth and Carpenter wanted to show something beyond sketches. So they built highly detailed models of some of the most interesting areas of the moon, complete with all their mountains, craters and rilles, then photographed them under the right lighting conditions for interesting shadows similar to what you'd see when that area was on the terminator.

I loved the idea, since I'd worked on a similar but much less ambitious project myself. About a decade ago, Dave North got the idea to make a 3-D model of the full moon that he could use for the SJAA astronomy class. I got drafted to help. We started by cutting a 3-foot disk of wood, on which we drew a carefully measured grid corresponding to the sections in Rukl's *Atlas of the Moon*. Then, section by section, we drew in the major features we wanted to incorporate. Once the drawing was done, we mixed up some spackle -- some light, and some with a little black paint in it for the mare areas -- and started building up relief on top of the features we'd sketched. The project was a lot of fun, and we use the moon model when giving talks (otherwise it hangs on the living room wall).

Nasmyth and Carpenter's models cover only small sections of the moon -- Copernicus, Plato, the Apennines -- but in amazing detail. Looking at their photos really is like looking at the moon at high magnification on a night of great seeing.



So I had to get the book. Amazon has two versions, a paperback and a hardcover. I opted for the paperback, which turns out to be scanned from a library book (there's even a scan of the pocket where the book's index card goes). Some of the scanning is good, but some of the plates come out all black. Not very satisfying.

But once I realized that an 1885 book was old enough to be public domain, I checked the web. I found two versions: one at Archive.org and one on Google Books. They're scans from two different libraries; the Archive.org scan is better, but the epub version I downloaded for my ebook reader has some garbled text and a few key plates, like Clavius, missing. The Google version is a much worse scan and I couldn't figure out if they had an epub version.

they mention, you'll have a great start on a lunar observing program that'll keep you busy for months. For experienced observers, it might give you a new appreciation of some lunar regions you thought you already knew well. Not at super-fine levels of detail -- no Alpine Valley rille -- but a lot of good discussion of each area.

Other parts of the book are interesting only from a historical perspective. The physical nature of lunar features wasn't a settled

issue in 1885, but Nasmyth and Carpenter feel confident that all of the major features can be explained as volcanism. Lunar craters are the calderas of enormous volcanoes; mountain ranges are volcanic too, built up from long cracks in the moon's crust, like the Cascades range in the Pacific Northwest.

There's a whole chapter on "Cracks and Radiating Streaks", including a wonderful plate of a glass ball with cracks, caused by deformation, radiating from a single point. They actually did the experiment: they filled a glass globe with water and sealed it, then "plunged it into a warm bath". The cracks that resulted really do look a bit like Tycho's rays (if you don't look TOO closely: lunar rays actually line up with the edges of the crater, not the center).

It's fun to read all the arguments that are plausible, well reasoned - - and dead wrong. The idea that

craters are caused by meteorite impacts apparently hadn't even been suggested at the time.

Anyway, I enjoyed the book and would definitely recommend it. The plates and observing advice can hold their own against any modern observing book, and the rest ... is a fun historical note.

There are a few things to look at in July besides the moon. Venus returns to the evening sky, though it stays low this month. Saturn is higher, a little past the meridian as night falls, so it's observable most of the evening.

(continued next page)

(Shallow Sky continued)

Uranus and Neptune are both available to late night observers: Neptune rises between 8 and 9, Uranus not til around midnight. Pluto is transiting around midnight and so observable all night. It's only about 33 degrees up at its highest, though, so it may be a tough fight locating it. In the morning twilight sky are Mars and

a telescope! But, failing that, the next best thing is NASA's lunar monitoring program: they keep cameras trained on the moon to detect impacts. They've seen hundreds of small impacts, most so faint enough they'd be difficult to see visually through a telescope.



Jupiter, which come together for a nice conjunction late in the month. They're closest on the morning of the 22nd, less than half a degree apart. Worth a look if you're rising early on that Monday morning.

But that's not very many planets. So let me close with one more bit of recent lunar news.

Of course, these days we know that lunar craters are created by meteorite impacts. That's something every lunar observer would love to see live through

But on March 17 they saw the brightest impact yet, at 4th magnitude. A star that bright is easily visible with the naked eye -- but I'm not sure how obvious it would have been on the illuminated surface of the moon. Probably not very. It sure would have been interesting in a telescope, though! Or if a similarly large impact happened on the shadowed part of the moon.

NASA figures the object was probably about 40 kg (88 lbs) and a little bigger than a basketball, traveling about 56,000 mph. It was most likely part of a swarm of meteors: the Earth was hit by quite a few meteors at about that time,

moving with the same general speed and direction. That's actually the motivation for the lunar monitoring program: they hope to learn more about near-earth asteroids that might impact Earth.

Of course, reading about a lunar impact on a NASA site isn't the same as seeing one for yourself through the eyepiece. But it's good to know someone is keeping an eye on these things.

—

Note: Write to Akkana at Akkana@shallowsky.com

Starry, Starry Night Unukalhai

Paul Kohlmler

The name Unukalhai looks like it could be Hawaiian. But it is not pronounced "oo-noo-al-high" but "oo-nook-ul-high" and it is, like most stars, an Arabic word. It means "the serpent's neck". At magnitude 2.65 you can see it naked eye but it seemed to me to be rather dim. It is also called Alpha Serpentis so it is the brightest star in the septent. The constellation Serpens is actually split into two pieces with Ophiuchus in the middle. Unukalhai is in the head "Serpens Caput" while the rest is in "Serpens Cauda", the Serpent's Tail.



However you remember the stellar spectra, OBAFGKM, the stellar classification K2 tells you the surface is cooler than our G class sun. But this orange giant has a diameter 15 times Sol. This means that if the sun was the same 73 light-years distant it would be only 2 to 3% as bright as Unukalhai. The dark gaps in this star's spectrum are particularly distinct indicating additional metals.

If you find Unukalhai you are close to a number of globular clusters including M5, M10, M12, M14 and M13 in Hercules. On July 4th Unukalhai transits the meridian at about 10 p.m.

Mooning At Houge Park - Lunar Targets for July 12

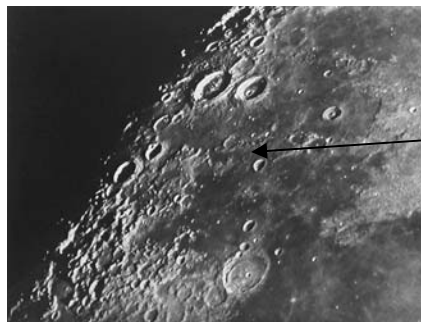
By Mark Wagner

Here are four targets that should be reasonably well placed for observing. It would be helpful to find someone with a Rukl Lunar Atlas, or to have a lunar map for reference. Coordinates listed if available. Images are courtesy of the Lunar and Planetary Institute.



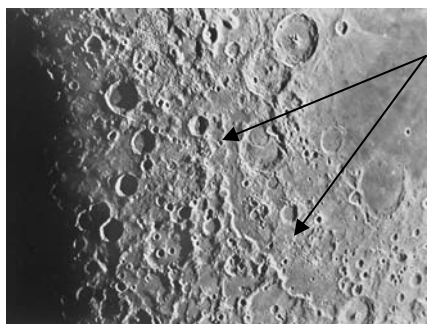
Aristotlis - Rukl pg 5.
50.2N, 17.4E

"The field around Aristotle is peppered with fine craters, much like Copernicus but without the obvious linearity ... they seem more random. Still it's a lovely field when



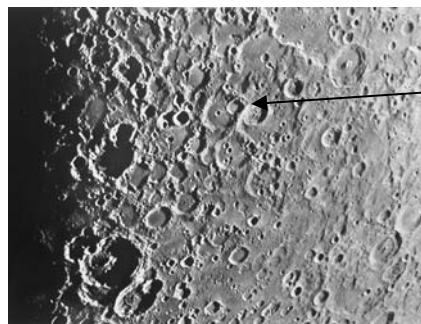
Lacus Somniorum - Rukl pg 14.

"Irregular contours, indefinite borders, surface area about . 70,000 sq km." - Astronomy Atlas of the Moon - Antonin Rukl.



Rupes Altai - Rukl pg 57
24S, 23E

"This mountain range appears to be a fault scarp sloping toward Mare Nectaris. It is quite prominent a few days after its sunrise." - Jay Reynolds Freeman. 480 km long.



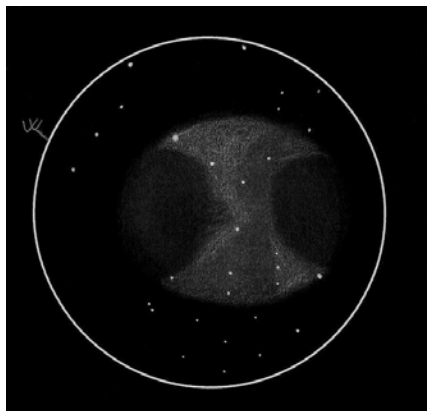
Rabbi Levi - Rukl pg 67
34.7S, 23.6E

A crater with worn sloped scalloped walls, and a series of craterlets arcing across its floor.. 81 km diameter.

Houge Park's Top Targets Of The Deep for July 14

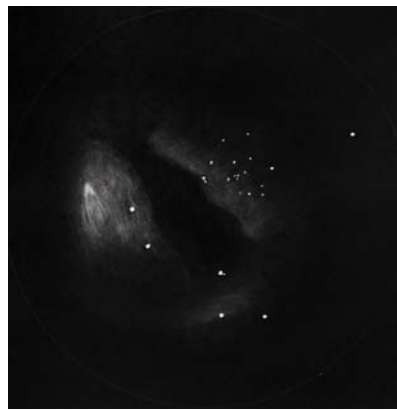
By Mark Wagner

The moon sets early at this Houge Star Party, so here are some deep sky targets. Even in brighter suburban skies, one can glimpse deep sky targets and have satisfying views. Drawings from Astronomy Sketch of the Day: <http://www.asod.info/>



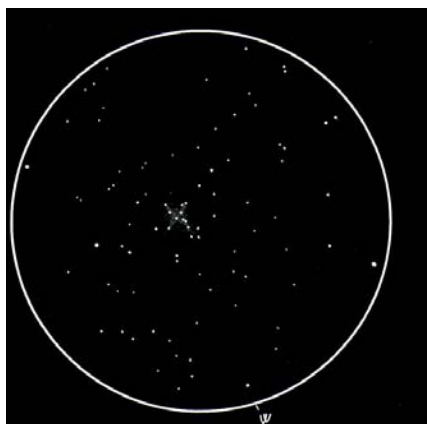
M27 Planetary nebula.
Vulpecula.
RA 20h 00m 09s
Dec +22°45'18"
Mag 8.10

Using a narrow band filter, such as a UCH or Ultrablock, will bring out more of the apple-core shape. Look for the central star.



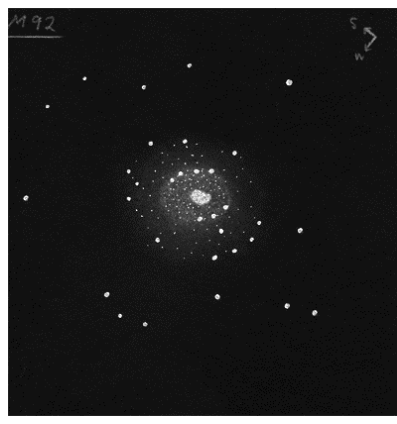
M8 Nebula and open cluster .
Sagittarius.
RA 18h 04m 36s
Dec -24°22'49"
Mag 5.80

This can be glimpsed without optical aid as a dim glow above the spout of the Teapot in Sagittarius. Use a narrow-band filter to enhance detail in the nebulosity.



M7 Open cluster.
Scorpius.
RA 17h 54m 46s
Dec -34°48'59"
Mag 3.30

This is a naked-eye target, even from Houge Park! A wide-field eyepiece will nicely frame this very bright, very large open cluster.



M92 Globular Cluster .
Hercules.
RA 17h 17m 29s
Dec +43°07'08"
Mag 6.50

Often overlooked due to its proximity to the Great Hercules Cluster (M13), M92 is well placed and a buzz of activity. Jump back to M13 and compare the two giant globulars.

A Few Board Reports

submitted at press time.

The Advanced Loaner Program is moving along just fine. The most significant improvement to happen was the clubs purchase of Rich's XT8g. This fully go to dob is an outstanding addition to the Advanced Loaner program and will be in great demand (assuming that I can get Greg to return it one of these days – lol). The XT8g is the perfect size as it has just enough aperture to see deep sky objects and yet remain portable for the majority of the club members. What makes this scope really cool is that our members will be able to use their iPhone, iPad, or Android device to connect to it and control it from the Sky Safari or Star Seeker software. We will include the required cables to allow them to do so.

Quick STARt program: Everything is going quite well for the most part. I have been opening up the program to those who would like to participate (sit in) using their own scope. This provides them with the basics that are taught as well as guidance in using their own scope (regardless of design). Plus these folks join us out at RCDO to not only learn how to star hop but to meet others and socialize. This is a huge part of the success of this program that needs to be recognized.

The program is taught once per month, on the 1st Friday following the 1st Sunday of the month. You can see the current status of all scopes/mounts in the Advanced Loaner program here: <http://www.sjaa.net/loaners.shtml>

Binocular Stargazing Program: Ed Wong came up with an idea to create this some time ago. We chatted about it for a number of months and then decided to move forward with it. We have held two sessions so far with the first being a practice run (given to other OSA docents, maybe 10 or so). Our 1st official session held on June 8th had close to 30 folks show up.

We hold these sessions at Rancho due to the dark skies and the fact that most folks have 50mm binoculars (so darks skies are needed – if not a must).

This program is generating an interest in the night sky that goes beyond binoculars – folks learn about SJAA and all of it's programs. Quite a few folks have joined SJAA as a result of this program. See info at: <http://www.meetup.com/OSA-Hiking-Enthusiasts/events/124523932/>

HEARD AND OVERHEARD... By Mina Wagner



I hear everybody is very busy getting ready for GSSP. Are you going?

I hear we are very close to having our brand new screen installed in Hogue Park. We will have it for sure for next meeting in July. We will then have a few movie nights. People have voted for which movie they would like to see. Did you get to vote?

Start preparing for July 19th. Make sure you make an appointment with your hairdresser, your beautician, your favorite tailor, for the following event:

Date: Wed, Jun 19, 2013 at 3:35 AM

Subject: JULY 19, 2013: A DAY TO CELEBRATE THE PALE BLUE DOT

June 18, 2013

Dear Friends and Colleagues,

Something great, something big, something very special that's never happened before is about to happen!

On July 19, 2013, NASA's Cassini spacecraft will be turned to image Saturn and its entire ring system during a total eclipse of the sun, as it has done twice before during its previous 9 years in orbit.

But this time will be very different. This time, the images to be collected will capture, in natural color, a glimpse of our own planet next to Saturn and its rings on a day that will be the first time Earthlings know in advance their picture will be taken from a billion miles away.

It will be a day for people all over the globe to celebrate together the extraordinary achievements that have made such an interplanetary photo session possible. And it will be a day to celebrate life on the Pale Blue Dot.



A new Captain's Log that expresses the significance of this event and how you can participate in it can be found in the usual place on <http://ciclops.org>

You can also find graphics showing where Earth will be relative to Saturn and what portion of the Earth will be illuminated during the picture-taking event at ...

http://ciclops.org/view_event/193/

Our planet, of course, will only be a blue point of light as seen from Saturn. But it, and we, are unique in all the solar system, and that is worth celebrating!

Enjoy!

Carolyn Porco
Cassini Imaging Team Leader

Upcoming SJAA Speakers and Events

August 17, General Meeting



Dr. Christopher P. McKay of NASA/Ames
His topic: The search for life on other planets, with an update from the Mars Curiosity Rover. See <http://spacescience.arc.nasa.gov/staff/chris-mckay> for more info on Dr. McKay.



Charlie Wicks and home-built refractor frequently attend CalStar

September 21, General Meeting



Our speaker: Mr. David Raimondi, LLNL, President of LUNAR (Livermore Unit of the National Association of Rocketry). His topic "Rocketry in Education".

CalStar shines October 3, 4 & 5! Can't make it to GSSP? Or even if you go, put CalStar on your calendar. Held at Lake San Antonio, good dark skies are a simple 2-1/2 hour drive from San Jose!

CalStar is for many, the favorite annual event, as it is very laid back and attracts astronomers together from both northern and southern California. The SJAA is integral in making this event happen, acting as a co-sponsor with the host group (TAC) and the Astronomical Association of Northern California. Watch for more information in the Ephemeris.

Also watch for information on TAC's webpage: <http://www.observers.org/>

2013 Astronomical Swap Meet

November 17 is the date for this year's SJAA fall swap. Never too early prepare. One person's "junk" is your treasure!

SJAA Board Meeting

No Board minutes. Meeting was on 6/28, after the Ephemeris went to press.

Publication Statement

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Articles for publication welcome. Must be submitted by the 16th of each month.

San Jose Astronomical Association
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San Jose Astronomical Association Membership Form

P.O.Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

- Membership Type:**
- ☐ Regular — \$20
 - ☐ Regular with Sky & Telescope — \$53
 - ☐ Junior (under 18) — \$10
 - ☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at**
<http://ephemeris.sjaa.net>.

Questions? Send e-mail
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Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to “SJAA”, or join/renew at <http://www.sjaa.net/membership.shtml>

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